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IE 58006 (Heuristic Optimization)

Assignment #1

Due Date: 08.11.2026 (until 23:59)

Problem Instance: 20-City Euclidean TSP

A solution is a permutation of the 20 cities. The tour is closed by returning from the last city to the city 0. Use Euclidean distances rounded to two decimal places:

$$d_{ij} = round\left(\sqrt{((x_i - x_j)^2 + (y_i - y_j)^2)}, 2\right)$$

City	x	y	City	x	y
0	0.5	0.5	10	6.8	0.7
1	2.0	0.8	11	8.5	1.1
2	4.2	0.5	12	9.2	3.4
3	5.4	2.0	13	8.1	5.2
4	5.0	4.0	14	6.3	5.5
5	3.2	5.1	15	1.1	7.0
6	1.0	4.7	16	3.6	7.4
7	0.2	3.0	17	6.0	7.2
8	2.3	2.2	18	8.8	7.0
9	3.8	2.8	19	9.7	5.5

Baseline initial tour: [0, 4, 1, 7, 3, 9, 2, 6, 8, 5, 15, 12, 18, 10, 16, 13, 11, 17, 14, 19, 0]

In your code, store each city only once; the final 0 is shown only to indicate the return to the starting city.

Implementation check: with the distance rule above, the length of this initial tour is **94.15**. Use this only to verify your distance/tour-length code.

Part A - SA Mechanics by Hand (20 points)

A1. 2-opt move evaluation and SA acceptance. **[12 points]**

From the baseline tour, consider a 2-opt move that removes edges (4,1) and (5,15), and adds edges (4,5) and (1,15). Use: $\Delta = d(4,5) + d(1,15) - d(4,1) - d(5,15)$

(a) Calculate Δ and state whether the candidate is improving or worsening.

(b) At $T = 3$, calculate the acceptance probability p for this move. ($p = e^{-\Delta/T}$)

(c) If $u = 0.65$, should the move be accepted at $T = 3$? Show the comparison between u and p.

(d) Repeat the decision for $T = 0.5$ using the same Δ and u. Briefly explain why the decision may change as temperature decreases.

A2. Cooling and best-so-far behavior. **[8 points]**

Use the lecture settings $T_0 = 3.0$, $\alpha = 0.90$, and reduce the temperature every 20 candidate moves using

$$T \leftarrow \alpha T$$

(a) Compute T immediately after 20, 100, and 200 candidate moves.

(b) Explain, in your own words, why the current tour length may increase during SA while the best-so-far tour length can never increase.

Part B - Implement Simulated Annealing (35 points)

Implement SA from scratch for the 20-city TSP. Your implementation must follow the mechanism used in the lecture notes.

B1. Solution representation and objective. **[5 points]**

- Represent a tour as a permutation of the 20 city indices.
- Compute the closed-tour length using the rounded Euclidean distance matrix.

B2. Random 2-opt neighbourhood. **[8 points]**

- Randomly select a valid 2-opt move.
- Evaluate the move using only the four affected edges:

$$\Delta = d(t_i, t_j) + d(t_{i+1}, t_{j+1}) - d(t_i, t_{i+1}) - d(t_j, t_{j+1})$$

- If the move is accepted, reverse the subpath between positions i+1 and j, as shown in the lecture notes.

B3. SA acceptance and cooling. **[8 points]**

- If $\Delta < 0$, accept the move.
- If $\Delta > 0$, compute $p = e^{-\Delta/T}$, generate u from $U(0,1)$, and accept if $u < p$.
- Use the baseline parameters $T_0 = 3.0$, $T_{min} = 0.02$, $\alpha = 0.90$, and reduce T every 20 candidate moves.
- Use random seed 7 for the baseline run.

B4. Current solution and best-so-far solution. **[6 points]**

- Use the current solution to generate the next neighbour; it may become better or worse.
- Maintain a separate best-so-far tour and update it only when the current tour improves on the best value.

B5. Required baseline-run outputs. **[8 points]**

- A table for the first 10 candidate moves containing: iteration, T, selected move indices, Δ, p and u when applicable, accepted/rejected, current length, and best-so-far length.
- The final best tour and its length.
- A plot showing current tour length and best-so-far tour length versus iteration, together with the temperature evolution.
- A visualization of the final best tour.

Part C - Controlled Computational Experiments (35 points)

Because SA is randomized, do not judge a configuration from a single lucky run. In all experiments below, use the same baseline initial tour and the same set of seeds so that comparisons are fair.

C1. Effect of the cooling factor α. **[15 points]**

Keep $T_0 = 3.0$, $T_{min} = 0.02$, and the cooling interval = 20 moves. Test $\alpha \in \{0.80, 0.90, 0.95, 0.99\}$ using seeds 0,1,...,9.

(a) For each α, report the best, mean, and standard deviation of the final best tour length over the 10 seeds.

(b) Report the number of candidate moves performed before T reaches T_{min}.

(c) Report the percentage of proposed worsening moves that were accepted.

(d) Explain the observed trade-off between cooling speed, diversification, solution quality, and computational effort.

C2. Effect of the initial temperature T_0. **[10 points]**

Return to the baseline $\alpha = 0.90$, $T_{min} = 0.02$, and cooling interval = 20 moves. Test $T_0 \in \{1.0, 3.0, 6.0\}$ using seeds 0,1,...,9.

Report the same statistics as in C1(a), the worsening-move acceptance percentage, and the number of candidate moves performed before T reaches T_{min}. Interpret the results in terms of high-/low-temperature behavior and computational effort.

C3. Recommended configuration. **[10 points]**

(a) Based on C1 and C2, choose one α and one T_0. Your choice should be justified by overall performance and robustness, not by one best run.

(b) Run the selected combination with seeds 0,1,...,9. Report best, mean, and standard deviation of the best tour length.

(c) Compare this combined configuration with the baseline settings. Did combining the individually preferred parameter values necessarily give the best overall behavior? Discuss briefly.

Part D - Interpretation and Reflection (10 points)

D1. Answer the following concisely, using evidence from your experiments. **[10 points]**

(a) Why does a high temperature encourage diversification, and why does a low temperature behave more like greedy local search?

(b) Why is it essential to store the best-so-far solution separately from the current solution?

(c) Give one example from your runs in which accepting a worsening move was useful for later improvement.

Submit the following

1. A concise PDF report containing your answers, tables, plots, and discussion.

2. Your complete source code with clear comments and reproducible random seeds.

3. Do not use a library that implements Simulated Annealing or solves the TSP for you.

4. Do not use an exact solver or a known optimal solution to guide or tune the SA search.

5. Clearly state any implementation choice not explicitly specified above.